

# E-COMMERCE CUSTOMER ANALYTICS AND CHURN PREDICTION

## PROJECT SYNOPSIS

### Project Title

E-Commerce Customer Analytics and Churn Prediction Using Machine Learning

### 1. Introduction

The E-Commerce Customer Analytics and Churn Prediction project is a data science and machine learning project designed to analyze customer purchasing behavior and identify customers who are likely to stop using an e-commerce platform.

The system analyzes historical customer transaction data and extracts useful information such as purchase frequency, total spending, recency of purchase, number of orders, and customer activity. These characteristics are used to build a machine learning model capable of predicting customer churn.

### 2. Problem Statement

E-commerce companies generate large amounts of customer transaction data, but identifying customers who are becoming inactive can be difficult when done manually. This project aims to analyze customer behavior and predict customers who are likely to churn so that businesses can take preventive retention actions.

### 3. Objectives

1. To collect and preprocess e-commerce customer data.
2. To identify and handle missing, duplicate, and inconsistent data.
3. To perform exploratory data analysis.
4. To analyze customer purchasing patterns.
5. To calculate Recency, Frequency, Monetary value, and number of orders.
6. To define customer churn based on customer activity.
7. To identify factors associated with customer churn.
8. To train machine learning models for churn prediction.
9. To evaluate model performance.
10. To provide insights for improving customer retention.

### 4. Technologies Used

| Technology | Purpose                              |
|------------|--------------------------------------|
| Python     | Data processing and machine learning |
| Pandas     | Data manipulation and analysis       |

|                  |                                 |
|------------------|---------------------------------|
| NumPy            | Numerical operations            |
| Matplotlib       | Data visualization              |
| Seaborn          | Statistical visualization       |
| Scikit-learn     | Machine learning                |
| SQL              | Data querying and analysis      |
| Jupyter Notebook | Development and experimentation |
| CSV/Database     | Data storage                    |

## 5. Methodology

Data Collection → Data Cleaning → Exploratory Data Analysis → Feature Engineering → Churn Definition → Machine Learning → Model Evaluation → Customer Insights

The transaction data will first be cleaned by handling missing values, duplicates, incorrect data types, and inconsistent records. Exploratory analysis will then be performed to understand customer behavior. Customer-level features such as recency, frequency, monetary value, and order count will be calculated.

A churn definition will be created based on customer inactivity for a specified period. The prepared data will be divided into training and testing datasets, followed by model training and evaluation.

## 6. Expected Outcome

The expected outcome is a machine learning-based customer churn prediction system that classifies customers as likely to churn or not likely to churn and provides analytical insights into customer behavior.

## 7. Future Scope

- Real-time churn prediction
- Power BI/Tableau dashboard
- Deep learning models
- Customer recommendation system
- CRM integration
- Automatic personalized retention offers
- Deployment as a web application or API